

Is gaming stress or Relaxation? An HRV-Based Investigation of physiological responses in young adults

André Alesi^{a,b,d,*}, Kristina Klier^c, Benedict Herhaus^d, Klara Brixius^e, Ingo Froböse^a, Katja Petrowski^{d,1}, Matthias Wagner^{f,1}

^a Institute of Movement Therapy and Movement-oriented Prevention and Rehabilitation, German Sport University, Cologne, Germany

^b Institut für Schlaf und Regeneration GmbH, Schramberg, Germany

^c Faculty of Informatics, Institute of Technical Informatics, University of the Bundeswehr Munich, Germany

^d Medical Psychology & Medical Sociology, University Medical Center Mainz, Johannes Gutenberg University of Mainz, Mainz, Germany

^e Institute of Cardiology and Sports Medicine, Germany Sports University, Cologne, Germany

^f Faculty of Human Sciences, Institute of Sports Science, University of the Bundeswehr Munich, Germany

ARTICLE INFO

Keywords:

Heart rate variability (HRV)
Competitive gaming
Computer Game
Esport
Stress

ABSTRACT

This study investigates the impact of video gaming activities versus watching a nature film on heart rate (HR) and vagally-mediated heart rate variability (vmHRV) in healthy young men. Employing a randomized within-subject design, 31 male participants (average age: 23 years; BMI: 25.68; gaming experience: 8.69 years; daily gaming time: 1.96 h) were assigned to either start with playing video games for 120 min on two consecutive days, followed by watching a nature film for the same duration in the following week, or the reverse order. HR and vmHRV were measured via ECG before and after each activity. Results showed that Video gaming activities had no significant effect on HR and vmHRV. It is indicated that video gaming does not appear to have notable physiological impacts on the autonomic nervous system. Conversely, watching a nature film significantly reduced HR and increased vmHRV compared to playing video games. These findings suggest that watching nature films exerts a calming effect and enhances parasympathetic activity, potentially offering health benefits through stress reduction and relaxation. This study contributes important insights into the physiological effects of digital media and underscores the need for further research in this area.

Objective: Playing video games is associated with increased autonomic sympathetic nervous system activity, which may lead to a reduction in vagally-mediated HRV (vmHRV) and Heart Rate (HR). Hence the aim of this study was to investigate the effect of the vmHRV and HR in healthy young adults between playing a computer game and watching a nature film.

Methods: The study was designed as a randomized within subject design with $N = 31$ male players (age: $M = 23,00 \pm 3,53$ years, BMI: $M = 25,68 \pm 3,34$, time of experience: $M = 8,69 \pm 4,74$ years, and daily gaming time: $M = 1,96 \pm 1,32$ h). Group A started in week 1 with two consecutive days of 120 min gaming and in week 2 with two consecutive days of 120 min watching a film. Group B started in the opposite order. HR and HRV were measured with ECG before and after the intervention.

Results: The results demonstrated that playing videogames had no effect on HR ($t(55) = 1.80, p = 0.08$) and vmHRV measured by HF-HRV ($t(55) = -1.57, p = 0.12$) and RMSSD ($t(55) = -1.48, p = 0.15$). In contrast, watching a nature film led to a lower HR ($t(561.6) = -7.29, p = 0<.001$) and an increase vmHRV measured by HF-HRV ($t(561.4) = 4.02, p = 0<.001$) and RMSSD ($t(561.4) = 4.97, p = 0<.001$) compared to playing videogame.

Conclusions: These findings indicate that, among this group of male participants, playing video games did not seem to affect the autonomic nervous system in a manner that notably modified heart rate or vagal tone. Conversely, viewing a nature documentary evokes a sense of calmness and increased parasympathetic activity, which may promote overall health.

* Corresponding author at: Institute of Movement Therapy and Movement-oriented Prevention and Rehabilitation, German Sport University, Am Sportpark Müngersdorf 6, 50933 Cologne, Germany.

E-mail address: andre.alesi@stud.dshs-koeln.de (A. Alesi).

¹ Shared senior authorship.

1. Introduction

Playing video games have surged in popularity over the last 30 years, becoming a ubiquitous part of modern culture [1]. In fact, more than 80 % of U.S. respondents to the Digital-Media-Trends survey 2022 play video games, with half of smartphone owners gaming on their mobile device every single day [2]. In 2022 the revenue in the Video Games Segment worldwide had the size of US\$ 319.91bn and is expected to show an annual growth rate (CAGR 2023–2027) of 7.80 %, resulting in a projected market volume of US\$ 502.40bn by 2027 [3]. The rise in the popularity of video games can be traced to the advent of improved and readily available broadband as well as organized competitive video game play, otherwise known as e-sports [4].

The vast majority of gamers worldwide have a healthy gaming experience[5]. These benefits include cognitive skills development [6], combatting loneliness[7]and stress relief[8]. As noted in the Compensatory Model [9], gaming is a versatile coping strategy for many individuals, defined as the cognitive and behavioral efforts to manage specific external and/or internal demands that are perceived as taxing or overwhelming the person's resources[10].

However, previous research has also identified potential negative health impacts of gaming. Some of the commonly identified adverse health effects for professional and nonprofessional players are stress [11], sleep disturbances[12], vision problems[13], musculoskeletal pain [14], overuse injuries[15], metabolic disorders[16]or weight gain[17], and other behavioral problems (i.e., addiction, violence, aggression) [18,19]. The I-PACE model (Interaction of Person-Affect-Cognition-Execution) provides a theoretical framework that facilitates the integration of these findings, suggesting that a complex interplay of individual predispositions, emotional responses, cognitive processes, and executive functions contributes to the development and maintenance of problematic gaming behaviors [20].

Heart rate variability (HRV) has been frequently used to assess the physiological impacts of gaming, as it reflects the balance between the sympathetic (fight-or-flight) and parasympathetic (rest-and-digest) branches of the autonomic nervous system [21]. Vagal-mediated HRV (vmHRV), which includes parameters like the root mean square of successive differences (RMSSD) and high-frequency HRV (HF-HRV), specifically measures parasympathetic nervous system activity [22]. However, research on the effects of video gaming on HRV has produced mixed results. Studies have indicated that gaming can elevate stress levels, as evidenced by increased low-frequency (LF) to high-frequency (HF) HRV ratios, indicative of increased sympathetic activity [23,24]. Other studies, such as that by Weaver et al. [25], found no significant differences in physiological arousal between video gaming and passive activities such as watching a movie, indicating that the physiological responses to gaming may depend on the context[25].

Recent research in esports has shown that competitive gaming can lead to significant variability in HRV responses, depending on player roles and in-game situations. Welsh et al. [26] emphasized that individual differences play a significant role in HRV outcomes, with some players exhibiting heightened stress responses while others show resilience. The complex nature of gaming environments suggests that HRV responses cannot be generalized without considering the specific context, such as game type (violent vs. non-violent), player role, and competitive pressure[26].

Furthermore, current studies emphasize that traditional interpretations of HRV, such as the LF/HF ratio, may oversimplify the relationship between HRV and stress responses, as the LF component is influenced by both sympathetic and parasympathetic activity [27,28]. As a result, recent studies have called for more focused use of vmHRV parameters like RMSSD, which provide a more accurate measure of parasympathetic function[22].

Welsh et al. [26] also identified methodological inconsistencies in the existing literature, including varying recording times, unstandardized environmental conditions (e.g., light, sound), and differences in

game types. These inconsistencies hinder comparisons between studies and emphasize the need for more rigorous experimental designs. Future research should aim to standardize HRV measurement protocols and explore how environmental and contextual factors influence HRV responses during gaming [26]. To address these inconsistencies and focusing on the role of vmHRV in stress regulation, this study explores how competitive gaming and passive digital engagement impact stress responses.

Research Hypothesis (RH1): Playing a video game will result in a decrease in vagally mediated HRV (RMSSD, HF-HRV), indicating increased physiological stress, whereas watching a nature film will increase vmHRV, reflecting relaxation.

Given these gaps in the literature, this study aims to fill this void by employing a highly controlled experimental design, focusing on vmHRV parameters in an active gaming environment versus a passive nature film viewing scenario. This approach aims to provide a more reliable understanding of stress responses across these two different types of digital engagement.

These findings could contribute to a better understanding of how digital engagement influences autonomic stress responses and guide future interventions or recommendations for healthy gaming practices.

2. Materials and methods

2.1. Study participants

Thirty-One male players aged 18–37 (age: $M = 23.00 \pm 3.53$ years, BMI: $M = 25.68 \pm 3.34$, time of experience: $M = 8.69 \pm 4.74$ years, and daily gaming time: $M = 1.96 \pm 1.32$ h) took part in the study. Only males were considered for participation, as the menstrual cycle and use of oral contraceptive agents can affect sleep or HRV [25].

The study participants were recruited through online tendering and notice boards at the authors institution. After recruitment, study inclusion and exclusion criteria were checked in a standardized interview. Although the initial inclusion criteria allowed for participants aged between 18 and 65 years to provide recruitment flexibility, the final sample consisted of participants aged 18 to 37 years. Inclusion criteria were being a male playing the games *League of Legends* (Riot Games, 2009) or *Counter Strike: Global Offensive* (Valve, 2012) and not being a professional e-sport player, in the definition not to earn personal salary by playing video games. Exclusion criteria were any acute and/or chronic medical illness, mental disorders, or the use of medications that affect heart rate or the central nervous system, such as beta blockers. Additionally, participants who used perception-altering substances (e.g., certain pain relievers, recreational drugs) or medications that specifically alter cardiovascular function were excluded from the study. Rationale for Game Selection: *League of Legends* and *Counter Strike: Global Offensive* were chosen due to their popularity and their ability to generate high levels of cognitive engagement and competitive pressure, essential for inducing stress-related changes in heart rate variability (HRV). Both games require complex decision-making and teamwork, providing ideal conditions to measure physiological stress responses. Other game genres that do not require real-time decision-making or intense competitive engagement were excluded to maintain the study's focus on high-stress digital environments. The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the University of the Bundeswehr Munich, Germany (05/28/2022). All participants were informed of the inherent risks and benefits of the study before signing an informed consent form.

2.2. Experimental design

The study was designed as a randomized within subject design. The entire study was conducted over a 6-month period to allow for the recruitment and testing of all participants. However, each participant was individually assessed within a tightly controlled 2-week window.

During these 2 weeks, participants underwent the planned interventions and measurements. Therefore, interim health checks or long-term monitoring over the entire study duration were not necessary, as the health data of participants were collected solely during their specific assessment period. After the participants had agreed to the study, formalities had been clarified, explaining how to fill the daily log and answer all open questions. Then the participants were randomly divided into two groups. Group A starts in week 1 with two consecutive days of 120 min gaming (adc) and in week 2 with two consecutive days of 120 min watching a film (pdc). Group B started in the opposite order. The time difference between condition 1 and condition 2 was exact one week. To compare the HRV measurements from adc to pdc, and to ensure that the same activities are carried out during the day, participants were asked to wear the ECG-Sensor for 48 h each condition (see Fig. 1). The investigations were conducted between 8:00 PM and 11:00 PM. Participants were strictly instructed to refrain from consuming alcohol, caffeine, energy drinks, nicotine, and stimulating teas for 24 h prior to the experiment, to minimize any short-term effects on heart rate and heart rate variability. During the experiment, participants were allowed to drink, but only beverages that did not contain these substances. The participants could freely choose for adc between two video games (*Counter Strike: Global Offensive*; Valve, 2012 or *League of Legends*; Riot Games, 2009) for the study. The DVD *Earth* (Universal Film, 2007) and *Earth: One Amazing Day* (Universal Film, 2017) was used for the control condition (pdc). Participants were instructed to have their last meal until 7 pm and check their environment parameters 30 min before starting. Given that both light stimulation and posture can influence alertness in both conditions [29], dim light conditions were held constant < 30 lx, distance from the television monitor (an additional light

source) was controlled (1 m), subject were instructed to maintain a semi supine posture, and the volume was controlled to a speaker volume (< 80db). For both conditions a normal pc or laptop were used (see Fig. 2). Fig. 3. Fig. 4. Fig. 5..

2.3. HRV recording and analysis

HRV was recorded using a portable single-channel electrocardiograph (ECG) device (Bittium Corporation, Faros™180, Oulu, Finland). This sensor, a small portable rechargeable unit, was applied on the chest where it registered (and stored) the interbeat intervals measured from the R-wave maximum of a beat to the R-wave maximum of the next, with a sampling frequency of 1,000 Hz. However, potential measurement errors can occur due to movement artifacts, improper electrode placement, or poor skin contact. To minimize these issues, participants were instructed to maintain a semi-supine posture during data collection. HRV indices were analyzed using the Kubios™ HRV software (Biomedical Signal Analysis Group, Department of Applied Physics, University of Kuopio, Kuopio, Finland). All recorded ECGs (31 participants × 4 measurements = 124) were checked by experienced personnel beat-to-beat. Artefacts were corrected using the deletion method. The rate of losses (number of heartbeats marked as artifact or ectopic beats divided by all heartbeats detected, in%) was < 0.06 %. The dependent variables were analyzed in terms of frequency (low frequency [LF], high frequency [HF]) and time (beat-to-beat intervals [R-R], standard deviation of the mean of the qualified NN interval [SDNN]. The proportion of consecutive NN intervals with a difference greater than 50 ms [pNN50] and the root mean square difference of consecutive normal RR intervals [rMSSD] were also measured [30].

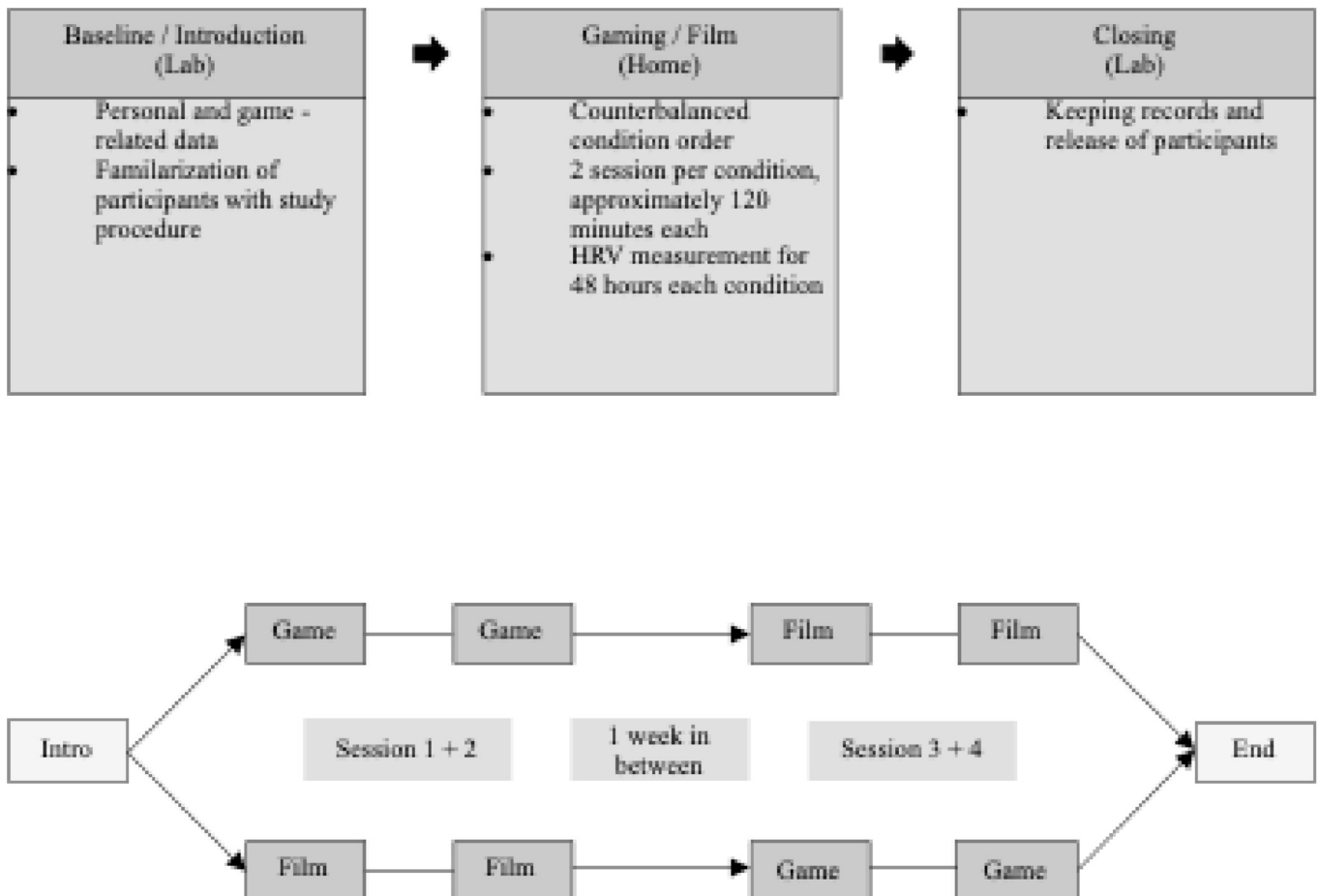


Fig. 1. Flow Chart of the Examined Study Protocol.

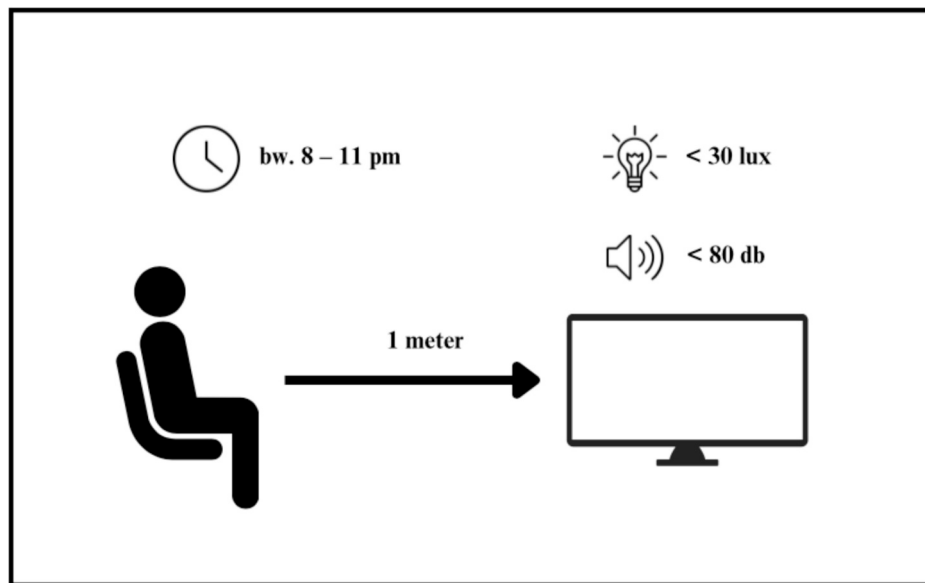


Fig. 2. Experimental setup during the gaming and film conditions, showing the participant's distance from the screen (1 m), controlled lighting conditions (<30 lx), sound level (<80 dB), and time window (between 8 and 11 PM).

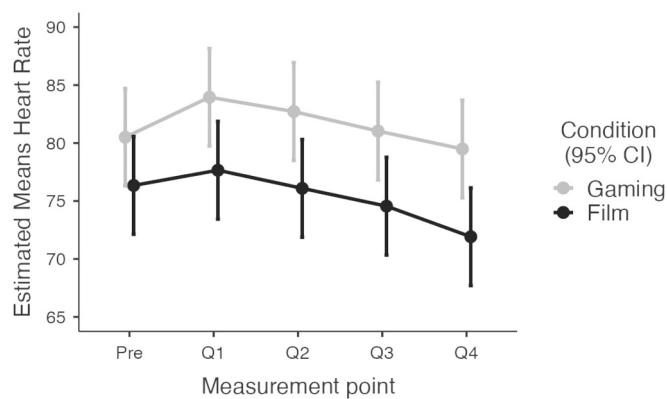


Fig. 3. Heart Rate responses to five measurement points. Pre was 5 min before the condition, Q1 was quarter one, Q2 was quarter two, Q3 was quarter three, Q4 was quarter four.

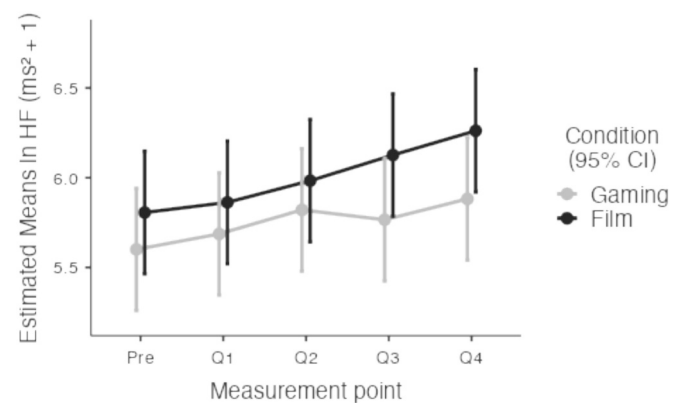


Fig. 5. HRV-HF responses to five measurement points. Pre was 5 min before the condition, Q1 was quarter one, Q2 was quarter two, Q3 was quarter three, Q4 was quarter four.

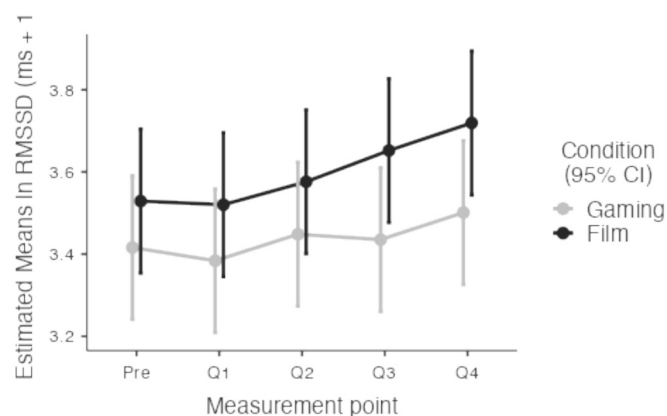


Fig. 4. RMSSD responses to five measurement points. Pre was 5 min before the condition, Q1 was quarter one, Q2 was quarter two, Q3 was quarter three, Q4 was quarter four.

2.4. Statistical analysis

The sample size was determined through power calculations conducted with G*Power v.3.1 [31]. Based on a medium effect size of Cohen's $f = 0.25$, two conditions (film vs. gaming), $n = 4$ repetitions, a significant level of $p = 0.05$ and power of 80 % ($1 - \beta = 0.80$), a total sample size of $N = 24$ for two-way ANOVA for repeated measures time x group interaction effect was needed.

For the statistical analysis, each heart rate, HRV-RMSSD and HF-HRV value was transformed by logarithm naturalis + 1 to obtain approximately normal distributions. First, baseline values of heart rate, HRV-RMSSD and HF-HRV of gaming and film condition were compared by using the dependent t -test. Second, the effect of the two conditions (gaming and film) to the heart rate and HRV parameters RMSSD and HF was analyzed using a mixed model with the interaction of measurement point Pre, Q1, Q2, Q3, and Q4 as a fixed effect. The intercept was treated as a random effect to account for intra-individual changes. The purpose of this analysis was to investigate whether the impact of condition on heart rate and HRV parameters varied across the five measurement time points. To test whether the testing day 1 and 2 of both conditions affects the heart rate and HRV parameters, fixed effect testing day was included

in the model. The software program used to analyze the data was *JAMOVI* [32].

3. Results

3.1. Heart rate

There was no significant difference in the baseline heart rate values between gaming and film condition ($t(55) = 1.80, p = 0.08$). Mixed model demonstrated a significant condition effect in the heart rate with lower values in the film condition compared to the gaming condition ($t(561.6) = -7.29, p \leq 0.001$). For measurement point Q4, a significant difference could be found with respect to measurement point Pre ($t(562.1) = -2.02, p \leq 0.05$), but there was no measurement point x condition interaction effect (see Table 1). With regard to two testing days of both condition, there was no effect of testing day in the heart rate.

Relation to RH1: The observed lower heart rate in the film condition supports RH1 by indicating a greater state of relaxation during passive digital engagement compared to gaming.

3.2. RMSSD

There was no significant difference in the baseline rMSSD values between gaming and film condition ($t(55) = -1.48, p = 0.15$). Mixed model demonstrated a significant condition effect in the RMSSD with higher values in the film condition compared to the gaming condition ($t(561.4) = 4.97, p \leq 0.001$). For measurement point Q4, a significant difference could be found with respect to measurement point Pre ($t(561.0) = 2.67, p \leq 0.05$), but there was no measurement point x condition interaction effect (see Table 2). With regard to two testing days of both condition, there was no effect of testing day in the heart rate.

Relation to RH1: The significantly higher RMSSD values in the film condition compared to the gaming condition are in line with RH1, indicating increased parasympathetic activity and relaxation.

3.3. HF-HRV

There was no significant difference in the baseline HF-HRV values between gaming and film condition ($t(55) = -1.57, p = 0.12$). Mixed model demonstrated a significant condition effect in the HRV-HF with higher values in the film condition compared to the gaming condition ($t(561.4) = 4.02, p \leq 0.001$). For measurement point Q2, Q3, Q4, a significant difference could be found with respect to measurement point Pre (Q2 ($t(561.0) = 1.98, p \leq 0.05$), Q3 ($t(561.0) = 2.41, p \leq 0.05$), Q4 ($t(561.0) = 3.66, p \leq 0.05$)), but there was no measurement point x condition interaction effect (see Table 3). With regard to two testing days of both condition, there was no effect of testing day in the heart rate.

Table 1

Mixed model results of the fixed effect parameter of interaction between Heart Rate (HR) and Film-Gaming Condition, the time points Q1, Q2, Q3, Q4, the testing day, and between the conditions (film/gaming) and the measurement points (Q1-Q4/pre).

Fixed Effects Parameter Estimates								
Names	Effect	Estimate	SE	95 % Confidence Interval		df	t	p
				Lower	Upper			
(Intercept)	(Intercept)	78.427	1.685	75.124	81.7290	30.0	46.545	<.001
Condition1	Film – Gaming	-6.227	0.855	-7.903	-4.5521	561.6	-7.285	<.001
Measurement point1	Q1 – Pre	2.385	1.344	-0.249	5.0201	561.0	1.774	0.077
Measurement point2	Q2 – Pre	0.985	1.347	-1.656	3.6254	561.1	0.731	0.465
Measurement point3	Q3 – Pre	-0.625	1.347	-3.266	2.0154	561.1	-0.464	0.643
Measurement point4	Q4 – Pre	-2.716	1.347	-5.357	-0.0756	561.1	-2.016	0.044
Testing Day1	2—1	-0.977	0.858	-2.659	0.7048	562.5	-1.139	0.255
Condition1 * Measurement point1	Film – Gaming * Q1 – Pre	-2.141	2.689	-7.410	3.1290	561.0	-0.796	0.426
Condition1 * Measurement point2	Film – Gaming * Q2 – Pre	-2.466	2.695	-7.747	2.8156	561.1	-0.915	0.361
Condition1 * Measurement point3	Film – Gaming * Q3 – Pre	-2.319	2.695	-7.601	2.9619	561.1	-0.861	0.390
Condition1 * Measurement point4	Film – Gaming * Q4 – Pre	-3.413	2.695	-8.694	1.8686	561.1	-1.267	0.206

Relation to RH1: The significantly higher HF-HRV values in the film condition compared to the gaming condition confirm RH1, suggesting enhanced vagal tone during the passive nature film viewing.

4. Discussion

The increasing popularity of video games has led to scientific scrutiny of possible unclear health effects [33]. On the one hand, gaming has been attributed with positive health benefits [34], while on the other hand, there are critical voices that oppose this notion [35]. Research into the physiological response to gaming, especially through HRV parameters, has been inconsistent, with limitations in methodology and lack of theoretical foundations highlighted in recent literature [26]. This study sought to investigate these effects in a controlled environment, comparing an active gaming condition (playing a violent video game) with a passive condition (watching a nature film).

Findings from the present study showed no significant differences in HR or vagally-mediated HRV (vmHRV) measured via HF-HRV and RMSSD, between the gaming and film-watching conditions. However, watching a nature film resulted in a reduction in HR and an increase in vmHRV, suggesting a shift towards greater parasympathetic activity and relaxation compared to playing video games.

Relation to RH1: These results are consistent with RH1, which proposed that video gaming would reduce vmHRV (indicative of stress), whereas film viewing would enhance vmHRV (indicative of relaxation). The observed physiological changes in vmHRV (RMSSD and HF-HRV) and heart rate support this hypothesis, indicating that passive digital engagement facilitates greater parasympathetic activation.

The findings of this study align with those reported by Weaver et al. [25], who observed that physiological arousal remained unchanged when comparing thirteen male adolescents in experimental (active video gaming) and control (passive DVD watching) conditions [25]. However, the current study did not replicate previous results from Higuchi et al. [36], who found significantly higher heart rates while playing video games compared to a control condition with a simple task with low mental load even 1.45 h later [36]. Additionally, the increase in LF/HF ratio and the decrease in RMSSD reported by Chaput et al. [23] and Porter et al. (2019) during video game play were not replicated in this study [23]; Porter et al., 2019).

The observed inconsistencies in physiological responses to video gaming across studies may be attributed to a variety of factors, including gender, duration of gameplay, level of immersion, individual gaming expertise, as well as potential influences from light exposure, sound, and food intake [37]. To mitigate these variables, the present study specifically included only young males to eliminate the influence of menstrual cycle fluctuations in females on vmHRV [38]. Additionally, to minimize potential confounders such as daily form or mood, the gaming duration was fixed at 2 h across two consecutive days. To ensure a standardized level of immersion in the game, the players' rankings at the start were

Table 2

Mixed model results of the fixed effect parameter of interaction between rMSSD and Film-Gaming Condition, the time points Q1, Q2, Q3, Q4, the testing day, and between the conditions (film/gaming) and the measurement points (Q1-Q4/pre).

Fixed Effects Parameter Estimates								
Names	Effect	Estimate	SE	95 % Confidence Interval		df	t	p
(Intercept)	(Intercept)	3.5180	0.0727	3.3756	3.6605	29.9	48.414	<.001
Condition1	Film – Gaming	0.1624	0.0327	0.0984	0.2264	561.4	4.972	<.001
Measurement point1	Q1 – Pre	−0.0206	0.0514	−0.1212	0.0801	560.9	−0.400	0.689
Measurement point2	Q2 – Pre	0.0396	0.0515	−0.0613	0.1405	561.0	0.769	0.442
Measurement point3	Q3 – Pre	0.0711	0.0515	−0.0298	0.1720	561.0	1.381	0.168
Measurement point4	Q4 – Pre	0.1375	0.0515	0.0366	0.2384	561.0	2.672	0.008
Testing Day1	2—1	−0.0158	0.0328	−0.0801	0.0485	562.1	−0.482	0.630
Condition1 * Measurement point1	Film – Gaming * Q1 – Pre	0.0234	0.1027	−0.1779	0.2248	560.9	0.228	0.820
Condition1 * Measurement point2	Film – Gaming * Q2 – Pre	0.0149	0.1030	−0.1869	0.2166	561.0	0.144	0.885
Condition1 * Measurement point3	Film – Gaming * Q3 – Pre	0.1039	0.1030	−0.0979	0.3056	561.0	1.009	0.314
Condition1 * Measurement point4	Film – Gaming * Q4 – Pre	0.1051	0.1030	−0.0967	0.3069	561.0	1.021	0.308

Table 3

Mixed model results of the fixed effect parameter of interaction between HF-HRV and Film-Gaming Condition, the time points Q1, Q2, Q3, Q4, the testing day, and between the conditions (film/gaming) and the measurement points (Q1-Q4/pre).

Fixed Effects Parameter Estimates								
Names	Effect	Estimate	SE	95 % Confidence Interval		df	t	p
(Intercept)	(Intercept)	5.8795	0.1416	5.60201	6.1570	29.9	41.525	<.001
Condition1	Film – Gaming	0.2567	0.0638	0.13167	0.3818	561.4	4.024	<.001
Measurement point1	Q1 – Pre	0.0714	0.1003	−0.12525	0.2680	560.9	0.712	0.477
Measurement point2	Q2 – Pre	0.1987	0.1006	0.00161	0.3958	561.0	1.976	0.049
Measurement point3	Q3 – Pre	0.2426	0.1006	0.04553	0.4397	561.0	2.413	0.016
Measurement point4	Q4 – Pre	0.3683	0.1006	0.17126	0.5654	561.0	3.663	<.001
Testing Day1	2—1	−0.0452	0.0640	−0.17071	0.0804	562.1	−0.705	0.481
Condition1 * Measurement point1	Film – Gaming * Q1 – Pre	−0.0297	0.2007	−0.42299	0.3636	560.9	−0.148	0.882
Condition1 * Measurement point2	Film – Gaming * Q2 – Pre	−0.0430	0.2011	−0.43712	0.3512	561.0	−0.214	0.831
Condition1 * Measurement point3	Film – Gaming * Q3 – Pre	0.1550	0.2011	−0.23918	0.5491	561.0	0.771	0.441
Condition1 * Measurement point4	Film – Gaming * Q4 – Pre	0.1741	0.2011	−0.22001	0.5683	561.0	0.866	0.387

used. An equalization of the entry level could lead to an under- or overestimation of the playing strength, which could have an impact on the measured results. Differences in individual gaming expertise might also explain the inconsistencies in the literature. In this study, however, all participants had several years of gaming experience in different video games and played on a daily basis. Environmental factors such as light and sound settings were uniformly controlled, alongside restrictions on caffeine intake and a specified window for the last meal before participation.

The content variation between video games and control conditions could also contribute to the disparate findings seen in previous research. In this study, participants playing interactive digital violent games (e.g., League of Legends, Counter Strike) were compared with those watching a nature movie. This setup revealed no significant changes in HR and vmHRV during the video game condition. Interestingly, when comparing two active digital gaming conditions — players of fighting video games (Mortal Kombat) versus players of cognitive video games (Tetris) — an increase in blood pressure and a decrease in RMSSD were observed in the fighting game condition[39]. Additionally, comparing a digital sports game (Fifa) with a resting condition revealed an elevated ratio between the low-frequency and high-frequency components of HRV in the sports game condition [23]. Consequently, playing video games in the present study may not have sufficiently stimulated the parasympathetic or sympathetic nervous system to induce relaxation or stress.

These observations give rise to the intriguing proposition that the act of playing video games, and not merely the act of playing itself, could exert a significant influence on physiological responses. While there was no discernible difference in physiological stress responses when comparing video games to watching a nature film, the differences between various types of video games indicate that the content and nature

of the game are critical factors. These findings underscore the complexity of video games' effects on physiological responses and highlight the need for a more nuanced consideration of video game content in future research.

The strength of this study is that the confounding effects of random variation (i.e. daily form, mood, and level of immersion) were minimized by having both groups perform the same condition in a standardized protocol on two consecutive days. Strict participation criteria (i.e. non-professional e-sport player, absence of sleep disorders, no diagnosed gambling disorder, and no consumption of perception-altering substances and/or pain relievers or cardiovascular-altering medications) were applied, and a high degree of standardization was chosen for the evening assessments (i.e. brightness and loudness level and digital medium viewing distance). Furthermore, an intricate and precise single-channel ECG analysis was employed to determine the HR and HRV parameters. The testing was conducted in the subject's own home environment, which minimizes the potentially confounding effects of laboratory settings.

However, several limitations of this study should be noted. Despite efforts to standardize the gaming experience by initiating participants at their individual score levels, comparing the respective game challenges across participants remains a complex endeavor. This approach, while aimed at equalizing the level of difficulty and engagement for each player, may not fully account for the dynamic nature of skill, focus, and adaptability that affects gameplay. Furthermore, the attempt to standardize subjects' daily activities encountered inherent limitations, potentially contributing to variations in the recorded cardiovascular parameters. These factors underscore the challenge of creating perfectly comparable conditions in gaming research, highlighting the nuanced influence of individual player experience and external lifestyle factors on physiological responses. Although strict participation criteria were

applied, unaccounted health-related factors, such as genetic predispositions, may have influenced the results. These aspects should be more carefully considered in future studies to increase the robustness of the findings. Additionally, it is important to note that the results may not be generalizable to female populations, as all participants were male.

In conclusion, the findings of this study suggest that playing violent video games, such as *League of Legends* or *Counter Strike*, does not activate the sympathetic nervous system or significantly alter HR or vmHRV in young men. These findings imply that moderate gaming, without intense emotional involvement, may not provoke a substantial stress response. In contrast, watching a nature film resulted in reduced HR and increased vmHRV, indicative of greater parasympathetic activity and relaxation.

Future research should explore the broader spectrum of video game genres and individual gaming characteristics to better understand the physiological effects of different types of digital engagement. Furthermore, studies should investigate how factors such as game immersion, competitive pressure, and player expertise contribute to stress responses, with a view to providing more targeted recommendations for promoting healthy gaming habits.

CRedit authorship contribution statement

André Alesi: Methodology, Conceptualization, Formal analysis, Writing – original draft, Investigation. **Kristina Klier:** Investigation, Methodology, Resources, Conceptualization. **Benedict Herhaus:** Software, Data curation, Visualization, Conceptualization, Methodology. **Klara Brixius:** Writing – review & editing. **Ingo Froböse:** Supervision. **Katja Petrowski:** Writing – review & editing, Conceptualization, Project administration, Supervision. **Matthias Wagner:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] A.J. Toth, N. Ramsbottom, M. Kowal, M.J. Campbell, Converging evidence supporting the Cognitive link between Exercise and Sport Performance: a dual Systematic Review, *Brain Sci.* 10 (11) (2020) 859, <https://doi.org/10.3390/brainsci10110859>.
- [2] Westcott, K., Arbanas, J., Arkenberg, C., Auxier, B., Loucks, J., & Downs, K. (2022, March). *2022 Digital media trends, 16th edition: Toward the metaverse*. Deloitte.com/insights.
- [3] Statista. (2023, March). *Video Games Worldwide*. <https://www.statista.com/outlook/dmo/digital-media/video-games/worldwide#revenue>.
- [4] Campbell, M. J., Toth, A. J., Moran, A. P., Kowal, M., & Exton, C. (2018). *eSports: A new window on neurocognitive expertise?* (pp. 161–174). doi: 10.1016/bs.pbr.2018.09.006.
- [5] J.H. Kristensen, S. Pallesen, D.L. King, M. Hysing, E.K. Erevik, Problematic gaming and sleep: a Systematic Review and Meta-Analysis, *Front. Psych.* 12 (2021), <https://doi.org/10.3389/fpsy.2021.675237>.
- [6] M. Barr, Video games can develop graduate skills in higher education students: a randomised trial, *Comput. Educ.* 113 (2017) 86–97, <https://doi.org/10.1016/j.compedu.2017.05.016>.
- [7] L.K. Kaye, R. Kowert, S. Quinn, The role of social identity and online social capital on psychosocial outcomes in MMO players, *Comput. Hum. Behav.* 74 (2017) 215–223, <https://doi.org/10.1016/j.chb.2017.04.030>.
- [8] L. Reinecke, Games and Recovery, *J. Media Psychol.* 21 (3) (2009) 126–142, <https://doi.org/10.1027/1864-1105.21.3.126>.
- [9] K. Koban, J. Biehl, J. Bornemeier, P. Ohler, Compensatory video gaming, gaming behaviours and adverse outcomes and the moderating role of stress, social interaction anxiety, and loneliness, *Behav. Inform. Technol.* 41 (13) (2022) 2727–2744, <https://doi.org/10.1080/0144929X.2021.1946154>.
- [10] S. Folkman, J.T. Moskowitz, Coping: pitfalls and Promise, *Annu. Rev. Psychol.* 55 (1) (2004) 745–774, <https://doi.org/10.1146/annurev.psych.55.090902.141456>.
- [11] M.J. Smith, P.D.J. Birch, D. Bright, Identifying Stressors and Coping strategies of Elite Esports competitors, *International Journal of Gaming and Computer-Mediated Simulations* 11 (2) (2019) 22–39, <https://doi.org/10.4018/IJGCMS.2019040102>.
- [12] S. Peracchia, G. Curcio, Exposure to video games: effects on sleep and on post-sleep cognitive abilities. a systematic review of experimental evidences, *Sleep Sci* 11 (4) (2018) 302–314, <https://doi.org/10.5935/1984-0063.20180046>.
- [13] C. Rechichi, G. De Moja, P. Aragona, Video Game Vision Syndrome: a New Clinical Picture in Children? *Journal of Pediatric Ophthalmology & Strabismus* 54 (6) (2017) 346–355, <https://doi.org/10.3928/01913913-20170510-01>.
- [14] H. Zwibel, J. DiFrancisco-Donoghue, A. DeFeo, S. Yao, An Osteopathic Physician's Approach to the Esports Athlete, *Journal of Osteopathic Medicine* 119 (11) (2019) 756–762, <https://doi.org/10.7556/jaoa.2019.125>.
- [15] J. DiFrancisco-Donoghue, J. Balentine, G. Schmidt, H. Zwibel, Managing the health of the eSport athlete: an integrated health management model, *BMJ Open Sport Exerc. Med.* 5 (1) (2019) e000467, <https://doi.org/10.1136/bmjsem-2018-000467>.
- [16] S.L. Calvert, A.E. Staiano, B.J. Bond, Electronic gaming and the Obesity Crisis, *New Dir. Child Adolesc. Dev.* 2013 (139) (2013) 51–57, <https://doi.org/10.1002/cad.20031>.
- [17] E.A. Vandewater, M. Shim, A.G. Caplovitz, Linking obesity and activity level with children's television and video game use, *J. Adolesc.* 27 (1) (2004) 71–85, <https://doi.org/10.1016/j.adolescence.2003.10.003>.
- [18] C.A. Anderson, B.J. Bushman, Effects of Violent Video Games on Aggressive Behavior, Aggressive Cognition, Aggressive Affect, Physiological Arousal, and Prosocial Behavior: a Meta-Analytic Review of the Scientific Literature, *Psychol. Sci.* 12 (5) (2001) 353–359, <https://doi.org/10.1111/1467-9280.00366>.
- [19] A.T. Prescott, J.D. Sargent, J.G. Hull, Metaanalysis of the relationship between violent video game play and physical aggression over time, *Proc. Natl. Acad. Sci.* 115 (40) (2018) 9882–9888, <https://doi.org/10.1073/pnas.1611617114>.
- [20] M. Brand, E. Wegmann, R. Stark, A. Müller, K. Wölfling, T.W. Robbins, M. N. Potenza, The Interaction of Person-Affect-Cognition-Execution (I-PACE) model for addictive behaviors: Update, generalization to addictive behaviors beyond internet-use disorders, and specification of the process character of addictive behaviors, *Neurosci. Biobehav. Rev.* 104 (2019) 1–10, <https://doi.org/10.1016/j.neubiorev.2019.06.032>.
- [21] L.K. McCorry, Physiology of the autonomic nervous system, *Am. J. Pharm. Educ.* 71 (4) (2007) 78, <https://doi.org/10.5688/aj710478>.
- [22] S. Laborde, E. Mosley, J.F. Thayer, Heart Rate Variability and Cardiac Vagal Tone in Psychophysiological Research - Recommendations for Experiment Planning, Data Analysis, and Data Reporting, *Front. Psychol.* 8 (2017) 213, <https://doi.org/10.3389/fpsyg.2017.00213>.
- [23] J.-P. Chaput, T. Visby, S. Nyby, L. Klingenberg, N.T. Gregersen, A. Tremblay, A. Astrup, A. Sjödin, Video game playing increases food intake in adolescents: a randomized crossover study, *Am. J. Clin. Nutr.* 93 (6) (2011) 1196–1203, <https://doi.org/10.3945/ajcn.110.008680>.
- [24] M. Ivarsson, M. Andersson, T. Åkerstedt, F. Lindblad, Playing a violent television game affects heart rate variability, *Acta Paediatr.* 98 (1) (2009) 166–172, <https://doi.org/10.1111/j.1651-2227.2008.01096.x>.
- [25] E. Weaver, M. Gradsar, H. Dohnt, N. Lovato, P. Douglas, The effect of presleep video-game playing on adolescent sleep, *Journal of Clinical Sleep Medicine : JCSM : Official Publication of the American Academy of Sleep Medicine* 6 (2) (2010) 184–189.
- [26] M.R. Welsh, E. Mosley, S. Laborde, M.C. Day, B.T. Sharpe, R.A. Burkill, P.D. J. Birch, The use of heart rate variability in esports: a systematic review, *Psychol. Sport Exerc.* 69 (2023) 102495, <https://doi.org/10.1016/j.psychsport.2023.102495>.
- [27] G.E. Billman, H.V. Huikuri, J. Sacha, K. Trimmel, An introduction to heart rate variability: methodological considerations and clinical applications, *Front. Physiol.* 6 (2015), <https://doi.org/10.3389/fphys.2015.00055>.
- [28] J.A.J. Heathers, Everything Hertz: methodological issues in short-term frequency-domain HRV, *Front. Physiol.* 5 (2014), <https://doi.org/10.3389/fphys.2014.00177>.
- [29] J.M. Zeitzer, D.J. Dijk, R. Kronauer, E. Brown, C. Czeisler, Sensitivity of the human circadian pacemaker to nocturnal light: melatonin phase resetting and suppression, *The Journal of Physiology*, 526 Pt 3 (Pt 3) (2000) 695–702, <https://doi.org/10.1111/j.1469-7793.2000.00695.x>.
- [30] B.W. Johnston, R. Barrett-Jolley, A. Krige, I.D. Welters, Heart rate variability: Measurement and emerging use in critical care medicine, *Journal of the Intensive Care Society* 21 (2) (2020) 148–157, <https://doi.org/10.1177/1751143719853744>.
- [31] F. Paul, E. Erdfelder, A. Buchner, A.-G. Lang, Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses, *Behav. Res. Methods* 41 (4) (2009) 1149–1160, <https://doi.org/10.3758/BRM.41.4.1149>.
- [32] Gallucci, M. (2019). *GAMLj: General analyses for linear models. jamovi module*. <https://gamlj.github.io>.
- [33] C.J. Ferguson, M. Coulson, J. Barnett, A meta-analysis of pathological gaming prevalence and comorbidity with mental health, academic and social problems, *J. Psychiatr. Res.* 45 (12) (2011) 1573–1578, <https://doi.org/10.1016/j.jpsychires.2011.09.005>.
- [34] N. Johannes, M. Vuorre, A.K. Przybylski, Video game play is positively correlated with well-being, *R. Soc. Open Sci.* 8 (2) (2021) rsos.202049, <https://doi.org/10.1098/rsos.202049>.

- [35] A. Drummond, J.D. Sauer, C.J. Ferguson, Do longitudinal studies support long-term relationships between aggressive game play and youth aggressive behaviour? a meta-analytic examination, *R. Soc. Open Sci.* 7 (7) (2020) 200373, <https://doi.org/10.1098/rsos.200373>.
- [36] S. Higuchi, Y. Motohashi, Y. Liu, A. Maeda, Effects of playing a computer game using a bright display on presleep physiological variables, sleep latency, slow wave sleep and REM sleep, *J. Sleep Res.* 14 (3) (2005) 267–273, <https://doi.org/10.1111/j.1365-2869.2005.00463.x>.
- [37] M.D. Griffiths, F. Nuyens, An Overview of Structural Characteristics in Problematic Video Game playing, *Curr. Addict. Rep.* 4 (3) (2017) 272–283, <https://doi.org/10.1007/s40429-017-0162-y>.
- [38] K.M. Schmalenberger, T.A. Eisenlohr-Moul, M.N. Jarczok, M. Eckstein, E. Schneider, I.G. Brenner, K. Duffy, S. Schweizer, J. Kiesner, J.F. Thayer, B. Ditzen, Menstrual Cycle changes in Vagally-Mediated Heart Rate Variability are Associated with Progesterone: evidence from two Within-Person Studies, *J. Clin. Med.* 9 (3) (2020) 617, <https://doi.org/10.3390/jcm9030617>.
- [39] A.M. Porter, P. Goolkasian, Video Games and stress: how stress Appraisals and Game Content Affect Cardiovascular and Emotion Outcomes, *Front. Psychol.* 10 (2019), <https://doi.org/10.3389/fpsyg.2019.00967>.